

System Design

This document details the visual UI design, user experience patterns, component hierarchy, page layouts, and operational mechanisms implemented across the **Hostel Outing Permission & Approval Management System**.

1. Design Goals

- Role-Driven Clarity:** Tailor every screen to the logged-in user's exact responsibilities (STUDENT, HOD, WARDEN, WATCHMAN).
 - Zero Training Requirement:** Provide intuitive cards, distinct call-to-action buttons, clear labels, and visual status indicators.
 - Safety & Audit Transparency:** Emphasize mandatory steps (such as Warden parent confirmation) and present full audit timelines.
 - Responsive Performance:** Ensure layout responsiveness across desktop monitors, tablets, and mobile screen viewports.
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2. User Experience

The user experience follows modern web design standards:

- **Color Palette:** Dark Slate background (slate-900/slate-800), crisp card surface containers, and vibrant role-based color accents (Indigo for Students, Purple for HODs, Emerald for Wardens, Amber/Teal for Watchmen).
 - **Feedback & Alerts:** Toast notifications and inline dismissible banners communicate action results (success, warning, error) instantly.
 - **Interactive Elements:** Micro-animations, smooth hover states, responsive cards, and clean modal dialog overlays.
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3. Role-Based Dashboards

The application provides dedicated dashboards mapped to each authenticated role:

- **Student Dashboard** (/student): Focused on outing creation, active request cards, status progress, and personal history inspection.
- **HOD Dashboard** (/hod): Focused on academic review of pending department requests with approval/rejection comments.
- **Warden Dashboard** (/warden): Focused on block-scoped requests, mandatory parent approval verification, and final clearance.
- **Watchman Gate Desk** (/watchman): High-efficiency desk for student search, directory browsing, gate exit logging, and gate return recording.

4. Navigation Structure

Navigation is managed via `Navbar.tsx` and `React Router DOM`:

- **Global Header:** Displays the system title, current user name, role badge, department/block tags, and a **Logout** button.
 - **Tab Navigation:** HODs and Wardens have quick toggle buttons between **Active Pending** requests and **Role History** view pages (`/hod/history` and `/warden/history`).
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5. Authentication UI

Page Component: `frontend/src/pages/LoginPage.tsx`

- **Visual Layout:** Centered card overlay on a dark gradient background with a prominent lock icon and system title.
 - **Features:**
 - Email & Password input fields.
 - Role switcher helper buttons for quick demo login credentials filling.
 - Demo Credentials Reference Table displaying sample logins for Student, HOD, Warden, and Watchman roles.
 - Quick link to the **Sign Up** page.
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6. Signup UI

Page Component: `frontend/src/pages/SignupPage.tsx`

- **Visual Layout:** Structured multi-column registration form with dynamic field switching based on selected role.
 - **Supported Roles for Registration:** STUDENT, HOD, WARDEN.
 - **Dynamic Field Dependencies:**
 - Selecting STUDENT: Requires Full Name, Email, Password, Confirm Password, Register Number, Department, Hostel Block, Academic Year, and Room Number.
 - Selecting HOD: Requires Department assignment (Hostel Block & Register Number hidden).
 - Selecting WARDEN: Requires Hostel Block assignment (Department & Register Number hidden).
 - **Validation:** Enforces password matching, valid email formatting, and required field completion.
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7. Student Dashboard

Page Component: frontend/src/pages/StudentDashboard.tsx

- **Header Section:** Displays student profile overview (Name, Register Number, Department, Hostel Block, Room Number).
 - **Outing Request Action Card:** Prominent + **New Outing Request** button opening the creation form:
 - Date Picker (outing_date): Minimum date set to today.
 - Time Pickers (leaving_time& expected_return_time).
 - Destination input & Reason textarea.
 - **Active Outings Section:** Grid of active outing cards displaying current status badges (PENDING_HOD, PENDING_WARDEN, APPROVED, EXITED).
 - **Cancellation Action:** Eligible pending requests (PENDING_HODor PENDING_WARDEN) feature a **Cancel Request** button.
 - **Personal History Button:** Opens a modal displaying historical outings and full audit timelines.
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8. HOD Dashboard

Page Component: frontend/src/pages/HodDashboard.tsx

- **Header Section:** Shows HOD profile, assigned department badge (CSEor ECE), and total pending request counter.
 - **Pending Outings List:** Displays pending requests (PENDING_HOD) submitted by department students:
 - Student card details: Name, Register Number, Academic Year, Hostel Block, Destination, Reason, Date, Leaving & Return times.
 - Optional HOD Comment input field.
 - **Approve Button:** Advances request to PENDING_WARDEN.
 - **Reject Button:** Terminates request with REJECTED.
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9. HOD History

Page Component: frontend/src/pages/HodHistory.tsx

- **Analytical Summary Cards:** Total Department Outings, Approved Count, Rejected Count, Late Returns Count.
 - **Controls & Filters:**
 - Search Bar: Filters by student name, register number, or outing ID #OUT-X.
 - Status Filter Dropdown (PENDING_HOD, PENDING_WARDEN, APPROVED, REJECTED, CANCELLED, EXITED, COMPLETED, LATE_RETURN).
 - Hostel Block Filter Dropdown.
 - **Outings Data Table / Cards:** Lists complete department history with audit timeline inspect buttons.
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10. Warden Dashboard

Page Component: frontend/src/pages/WardenDashboard.tsx

- **Header Section:** Shows Warden profile, assigned hostel block badge (A Block, B Block, or C Block), and pending count.
 - **Pending Requests List:** Displays HOD-approved requests (PENDING_WARDEN) for block residents.
 - **Mandatory Parent Confirmation UI:**
 - Interactive Checkbox: [] Parent / Guardian Approval Confirmed by Phone.
 - **Parent Contact Button:** Quick trigger to mark parent confirmation status (POST /api/warden/outings/{id}/parent-confirmation).
 - **Action Buttons:**
 - **Approve Outing:** Enabled only after parent confirmation checkbox is checked. Advances status to APPROVED.
 - **Reject Outing:** Rejects request with optional reason.
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11. Warden History

Page Component: frontend/src/pages/WardenHistory.tsx

- **Analytical Summary Cards:** Total Block Outings, Approved Count, Exited Count, Completed Count, Late Returns Count.
 - **Controls & Filters:** Search bar (name/ID), Status filter, Department filter.
 - **Outings Data Table:** Displays historical block records, parent confirmation timestamps, and late return delay indicators.
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12. Watchman Dashboard

Page Component: frontend/src/pages/WatchmanDashboard.tsx

- **Operational Layout:** High-efficiency dual-section layout for main gate security officers.
 - **Today's Active Outings Section:** Displays students currently authorized to exit or return:
 - Displays student avatar icon, Name, Register Number, Department, Hostel Block, Outing ID.
 - **Record Exit Button:** Enabled when status is APPROVED. Updates status to EXITED.
 - **Record Return Button:** Enabled when status is EXITED. Updates status to COMPLETED or LATE_RETURN.
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13. Student Directory

Integrated Component: WatchmanDashboard.tsx(Directory Tab)

- **Purpose:** Campus-wide student lookup desk for Watchman security verification.
 - **Features:**
 - Live Search Bar: Real-time search across student register numbers, names, and emails.
 - Department & Hostel Block Filter Dropdowns.
 - Student Directory Grid: Displays student cards with year, room number, current active outing status, and quick action buttons.
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14. Outing Details Modal

Component: frontend/src/components/OutingDetailModal.tsx

- **Overlay Window:** Displays complete outing metadata:
 - Student Information: Name, Register Number, Department, Hostel Block, Room Number.
 - Outing Information: Date, Leaving Time, Expected Return Time, Destination, Reason.
 - Current Status & Parent Approval Status.
 - **Audit Timeline Component:** Full audit log visualization.
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15. Audit Timeline

Component: frontend/src/components/Timeline.tsx

- **Visual Design:** Vertical chronological timeline with connected icon nodes.
 - **Timeline Nodes:**
 - SUBMITTED: Blue icon node.
 - HOD_APPROVED/ WARDEN_APPROVED: Green checkmark nodes.
 - PARENT_APPROVAL_CONFIRMED: Purple phone checkmark node.
 - EXIT_RECORDED: Amber gate node.
 - RETURN_RECORDED/ COMPLETED: Emerald arrival node.
 - LATE_RETURN_DETECTED: Orange warning node with delay minutes badge.
 - REJECTED/ CANCELLED: Red cross nodes.
 - **Node Content:** Action title, actor name, actor role badge, comment string, formatted timestamp.
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16. Status Badges

Component: frontend/src/components/StatusBadge.tsx

The system uses standard status pill badges:

Status Key	Badge Color Styling	Visual Label
PENDING_HOD	bg-amber-500/10 text-amber-400 border-amber-500/20	Pending HOD
PENDING_WARDEN	bg-purple-500/10 text-purple-400 border-purple-500/20	Pending Warden
PENDING_WARDEN_ASSIGNMENT	bg-purple-500/10 text-purple-400 border-purple-500/20	Pending Warden Assignment
APPROVED	bg-emerald-500/10 text-emerald-400 border-emerald-500/20	Approved

Status Key	Badge Color Styling	Visual Label
REJECTED	bg-rose-500/10 text-rose-400 border-rose-500/20	Rejected
CANCELLED	bg-slate-500/10 text-slate-400 border-slate-500/20	Cancelled
EXITED	bg-blue-500/10 text-blue-400 border-blue-500/20	Exited (Off Campus)
COMPLETED	bg-green-500/10 text-green-400 border-green-500/20	Completed
LATE_RETURN	bg-orange-500/10 text-orange-400 border-orange-500/20	Late Return

17. Parent Approval Confirmation

- **UX Design:** Prominent checkbox and action button in WardenDashboard.tsx.
 - **Enforcement:** Final **Approve Outing** button remains disabled or returns backend HTTP 400 until parent consent is explicitly confirmed.
 - **Audit Logging:** Appends a PARENT_APPROVAL_CONFIRMED event to the audit timeline.
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18. Gate Exit / Return UI

- **Watchman Interface:**
 - **Exit Action:** Highlighted blue **Record Exit** button for APPROVED outings.
 - **Return Action:** Highlighted emerald **Record Return** button for EXITED outings.
 - **Confirmation Feedback:** Toast alert notifies the Watchman upon successful recording. If return is late, a warning alert displays the delay minutes.
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19. Late-Return Presentation

- **Badge Styling:** High-contrast orange badge (LATE_RETURN).
 - **Delay Display:** Renders explicit delay duration in minutes (e.g. Late Return (Delay: 35 mins)).
 - **Timeline Detail:** Displays Expected Return, Actual Return, and computed delay minutes.
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20. Responsive Design

- Built with Tailwind CSS responsive grid utilities (grid-cols-1 md:grid-cols-2 lg:grid-cols-3).
 - Navigation bar collapses cleanly into standard mobile layouts.
 - Tables feature horizontal scroll containers (overflow-x-auto) to prevent layout breaking on mobile devices.
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